



JOHNS HOPKINS
CAREY BUSINESS SCHOOL

Final Review

BU.330.760 Generative AI for Business

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Peer Evaluation



» Do not forget to submit your peer evaluation by midnight



Time and Format

- » Length: 1 hour
- » Format: **in-person, closed-book, no formula sheet**
 - No score for final if no show in classroom
- » Software: LockDown Browser
 - <https://www.respondus.com/lockdown/download.php?id=123533816>
 - You will not be able to access the exam if you try to access it via any other browser (Chrome, Firefox, Safari, etc.)
 - Sample test under Module 7



Question Types

- » 10 multiple-choices (1pt each) —
- » 3 essay questions (5pts each) —
 - Details and complete sentences expected
- » No coding, no Python
- » All conceptual, discussion, or analysis-based
- » Focus on understanding



Extended Office Hours

- » Thursday, May 8th, 11:30am-1:30pm
 - » Friday, May 9th, 11:30am-1:30pm
 - » Saturday, May 10th, 5pm-6pm
 - » Sunday, May 11th, 4pm-6pm
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- » Please use office hour Zoom link
 - » Feel free to email me with questions

Scope



- » All class materials we covered including
- Lecture slides (except those have “Optional” in title)
 - Labs
 - Assignments





Introduction

- » AI vs Machine Learning vs Deep Learning vs Generative AI
- » Idea of generation—
- » Model types: text, image, multi-modality —
- » Use cases of generative AI —
 - Customer experience —
 - Productivity and process optimization —
 - Content creation —



Foundation Model

- » Definition of foundation model ✓
- » Model architecture ✓
- » Encoder-decoder architecture ✓
 - Two sub-networks
 - Embedding
- » Autoencoder vs Variational autoencoder ✓
 - Discrete vs Continuous



Transformer

- » Encoder-decoder with attention
- » Attention mechanism description
- » Query, key, value parameters and how do they work
 - Mathematical calculation is not required
- » Positional encoding
- » Benefits of attention
 - Parallelism: multi-head attention
 - Interpretability: attention score



Pre-training and Fine-tuning

- » Pre-training description —
- » Pre-training approaches
 - Encoder: BERT
 - Decoder: GPT
- » Two fine-tuning approaches
 - Instruction tuning
 - Reinforcement learning from human feedback
- » Temperature
 - Low vs high



GenAI Ecosystem



- » Hardware: GPU
 - FLOPS
- » Cloud platforms
- » Foundation models and data
- » Applications and services





LLM: language vs reasoning

- » Large data + large model —
- » Able to name a few key models —
- » Open-source vs proprietary —
- » Key techniques in developing reasoning capacities: —
 - Large-scale reinforcement learning
 - Chain of thought
- » Comparison of language models and reasoning models —
- » DeepSeek approach —
 - More reinforcement learning than supervised fine-tuning
 - Distillation



Prompt Engineering and Agentic AI

- » Able to name a few prompt elements —
- » Few-shot prompting compared to zero-shot —
- » Automation and agents —
- » Five patterns —
 - Reflection .
 - Tool use .
 - ReAct .
 - Planning .
 - Multi-agent .
- » Why IT is the new HR



BloombergGPT and RAG

» Four ways to use LLMs, and comparison —

- Train your own model ✓
- Fine-tune an existing model ✓
- RAG ✓
- Prompt engineering ✓

» RAG

- Two components ✓
- Vector database and similarity search ✓
- Why chunking ✓
- Challenges ✓

» Comparison between BloombergGPT approach and RAG }

LLM Risks



- » OpenAI's model spec
- » Anthropic's constitutional AI
- » Able to name a few vulnerabilities in LLM
- » Prompt injection/adversarial prompting
 - Direct and indirect
 - Jailbreak and encoding attack



Image and Multi-Modality

- » Basic idea of GAN
 - Two sub-networks
 - Training, only high level, and why difficult to train
 - Usages
- » Basic idea of Diffusion
 - Two processes: forward diffusion and reverse diffusion
- » Multimodal
 - Basic idea of text-to-image
- » No modeling details for image and multimodal

Sweet Relief Social



» Please stay after final exam for the “Sweet Relief Social”



Course Evaluation



Please leave your comments in final course evaluation

I will be able to use your input next time

Good luck with your future endeavors!

